MAROOF AHMED MEMON

IT Graduate | Python Developer | Data Analytics & Machine Learning

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PROFESSIONAL SUMMARY

Information Technology graduate with a CGPA of 3.46/4.00 and one year of professional experience in front-end web development. Familiar with Python programming, data analysis, machine learning fundamentals, SQL databases, REST APIs, and responsive web development. Hands-on project experience includes bird sound classification using convolutional neural networks, customer churn prediction, sales forecasting, business dashboards, and REST API development. Interested in applying software development and analytical skills to practical technology problems while continuing to strengthen expertise in Python, data analytics, and artificial intelligence.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript
- **Data Analysis:** Pandas, NumPy, Matplotlib, exploratory data analysis, data preprocessing
- **Machine Learning:** Scikit-learn, classification, feature engineering, model evaluation fundamentals
- **Deep Learning & Audio Processing:** TensorFlow, Keras, Librosa, convolutional neural networks (CNNs), spectrogram generation
- **Databases:** Oracle SQL, MySQL
- **Visualization:** Power BI, dashboard development
- **Web Development:** HTML, CSS, Django, REST APIs
- **Tools:** Git, GitHub, Postman, Visual Studio Code, Jupyter Notebook

PROFESSIONAL EXPERIENCE

Front-End Web Developer

OS Synergy Software House | February 2025 – March 2026

- Developed and maintained responsive web pages using HTML, CSS, and JavaScript.
- Worked on front-end layouts and user-interface components to support usability across different screen sizes.
- Identified and fixed layout, styling, and interface-related issues during development and testing.
- Used Git and GitHub for source-code management and collaboration.
- Supported code maintenance, implementation updates, and basic technical documentation.

EDUCATION

Bachelor of Science in Information Technology (BS IT)

University of Sindh, Jamshoro, Pakistan | 2020–2024

- **CGPA:** 3.46/4.00
- Relevant areas: Programming, Database Management Systems, Data Structures and Algorithms, Software Engineering, Data Science, and Computer Vision.

PROJECT DETAILS

1. DeepBird: AI-Driven Bird Sound Identification

Technologies: Python, Librosa, TensorFlow, Keras, CNN

- Developed a bird sound identification project that classifies audio recordings into bird-species categories using a deep-learning approach.
- Used Librosa to process audio and generate spectrogram representations for model training and evaluation.
- Contributed to training a convolutional neural network (CNN) to recognize patterns in bird audio representations.
- Worked with a dataset covering 20 bird species and reported approximately 91% classification accuracy.
- Collaborated with a three-member team, with primary responsibility for CNN model training.
- The project received **second position**, as reported for the project.

2. Customer Churn Prediction

Technologies: Python, Pandas, NumPy, Scikit-learn

- Developed a machine-learning workflow to explore customer data and predict potential customer churn.
- Used Pandas and NumPy to prepare and manipulate data for analysis.
- Performed exploratory data analysis to understand feature distributions and identify patterns relevant to churn.
- Applied data preprocessing and feature engineering to prepare input variables for predictive modelling.
- Used Scikit-learn to build and assess classification models as part of the prediction workflow.

3. Sales Forecasting and Business Dashboard

Technologies: Python, Scikit-learn, Power BI

- Worked with historical sales data to explore trends and support business-performance analysis.
- Prepared and analyzed data using Python as part of a sales forecasting workflow.
- Used Scikit-learn to develop a machine-learning-based forecasting workflow.
- Created a Power BI dashboard to present sales information and make business data easier to review.
- Combined predictive analysis and dashboard visualization to support data-driven interpretation of sales performance.

4. Employee Management REST API

Technologies: Python, Django REST Framework, Postman

- Developed a REST API for managing employee-related records.
- Implemented CRUD operations to create, retrieve, update, and delete employee information.
- Used JSON-based requests and responses to support communication between clients and the API.
- Tested API endpoints using Postman to check request handling and response behaviour.
- Organized the API functionality around standard REST principles and structured endpoint interactions.

COMPETITIONS & RECOGNITION

- **DeepBird Project — 2nd Position:** Trained the CNN model with a three-member team and presented the project to the jury; awarded second position.
- **Hackmakers World Innovation Day Hack 2022:** Collaborated with an international team across time zones and presented our solution to the judging panel.
- **Hackmakers FormulaAI Hack 2022:** Worked with a diverse global team to research, prototype, and present an AI-based solution under a tight deadline.
- **Meta Hacker Cup 2023:** Solved real-world problems built around data structures and algorithms under competitive, time-pressured conditions.
- **NanotechX:** Participated in the technology-focused event.